Optical Predictability at Scale:

Expanded Beam Technology for Next-Generation AI Networks

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Working Group	SROI (Short Reach Optical Interface)
Title	*Optical Predictability at Scale: Expanded Beam Technology for Next-Generation AI Networks*
Source	Rebecca Lovely, Josh Wilken, and Michael Ransford
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Abstract

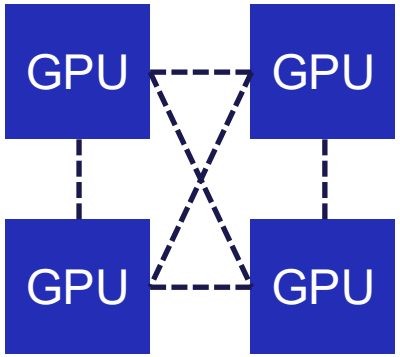
A discussion of lens-based optical interface architectures for next-generation AI and hyperscale data center interconnects. The presentation explores how expanded-beam optical connectivity implemented in the MMC with Prizm® TMT Technology may address challenges associated with density scaling, contamination tolerance, operational predictability, and deployment efficiency in front-plane and back-plane applications. Considerations for reliability, field readiness, and system-level performance are also discussed.

What We'll Cover

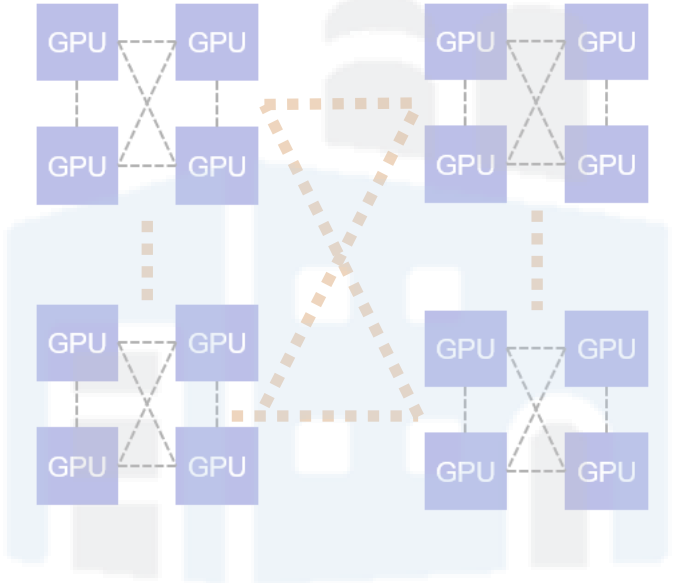
- Barriers to Growth: Physical and Operational Limitations
- Why Prizm? Environmental resiliency and remate stability guiding optical predictability
- Density: More channels, more connections using the same VSFF footprint
- Termination Advantage: Technology that enables manufacturing scale for termination
- Where we go next: Reduced insertion loss

Scale-Up: The Next Major Frontier for Optical Connectivity

Scale Up



Scale Out



Clusters

Scale Across



AI Factory

Compute Node

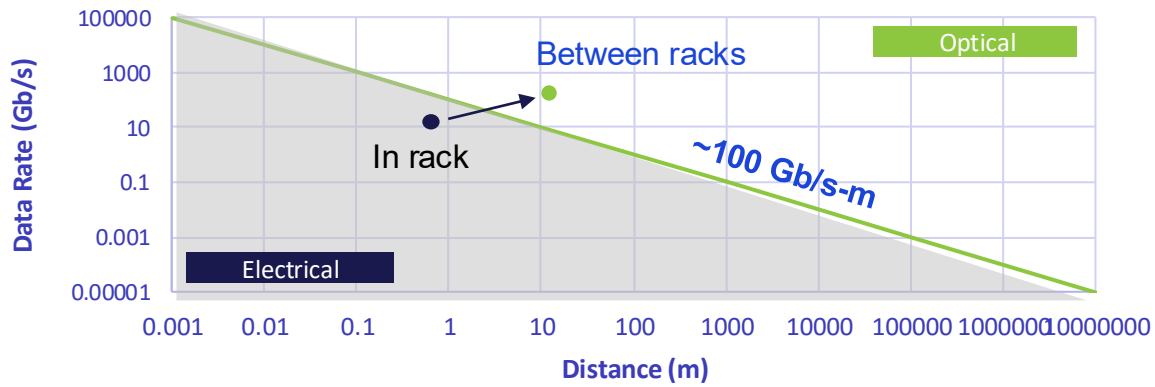
Emerging Optical conversion from copper

As node sizes increase, copper becomes the constraint

AI models grow **~100x every 2 years**, requiring more interconnected XPUs

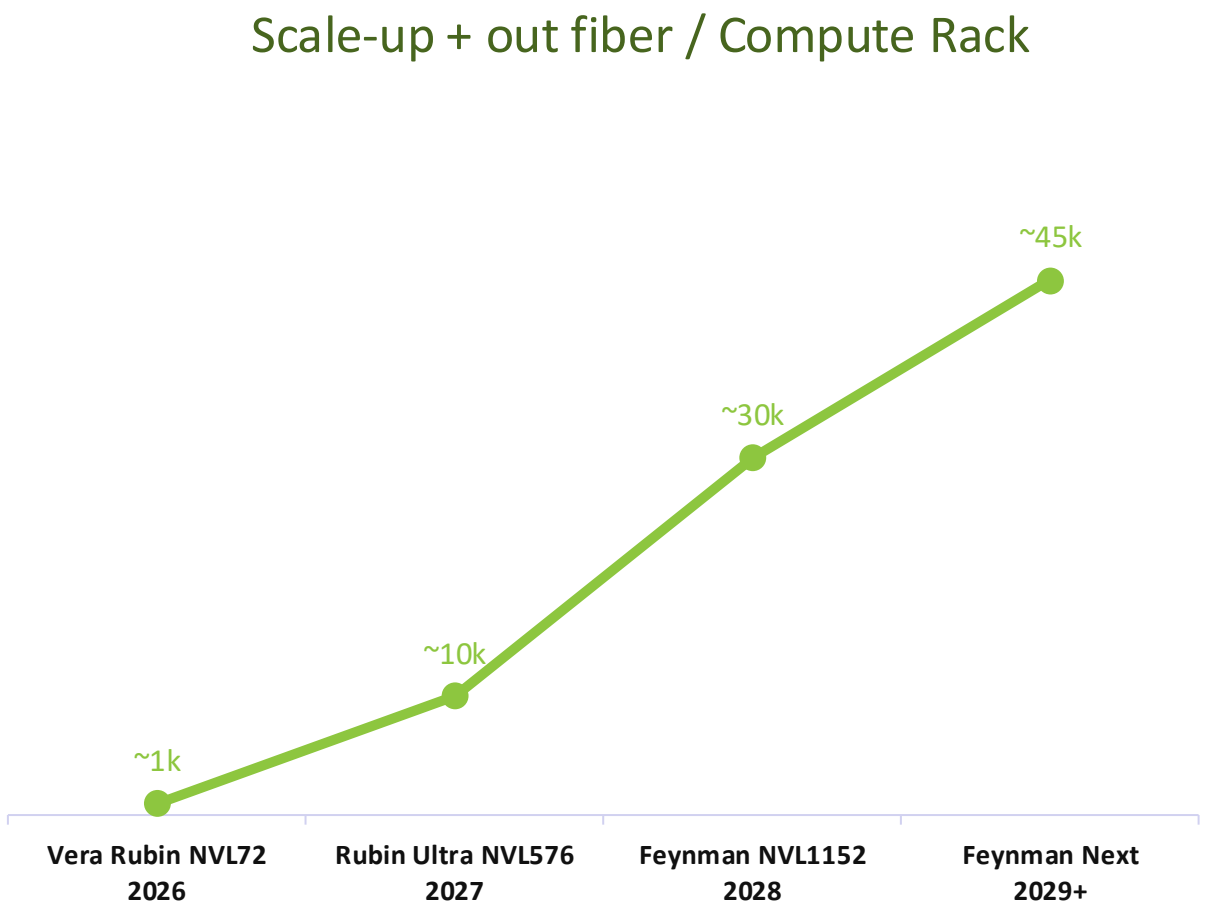
As node sizes grow beyond 72 XPUs, copper reach x speed constrained

Data Rate and Distance Requirements, Scale Up

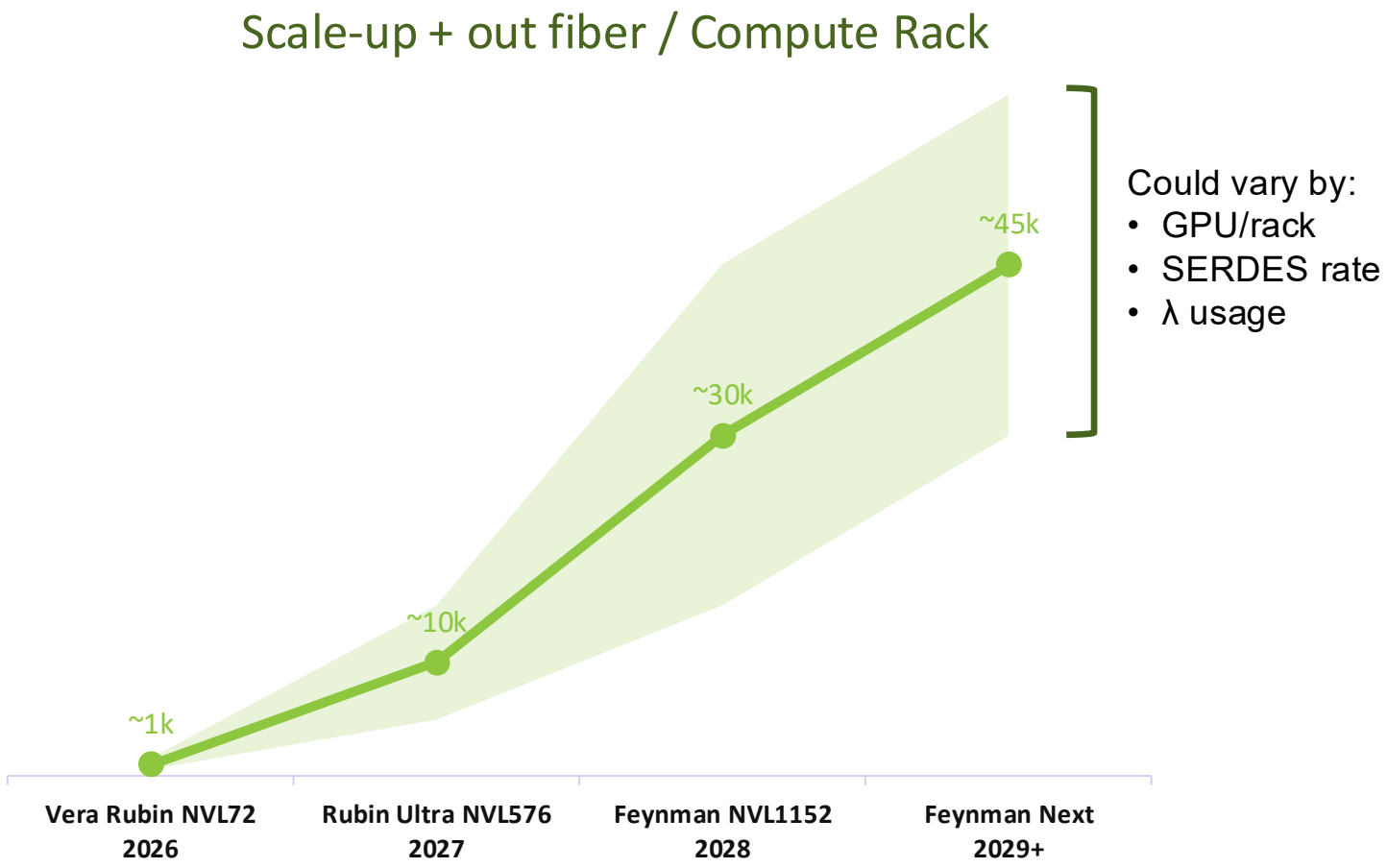


Optical solutions solve for reach at higher data rates

Scale-up drives rapid in-rack fiber growth in coming years:

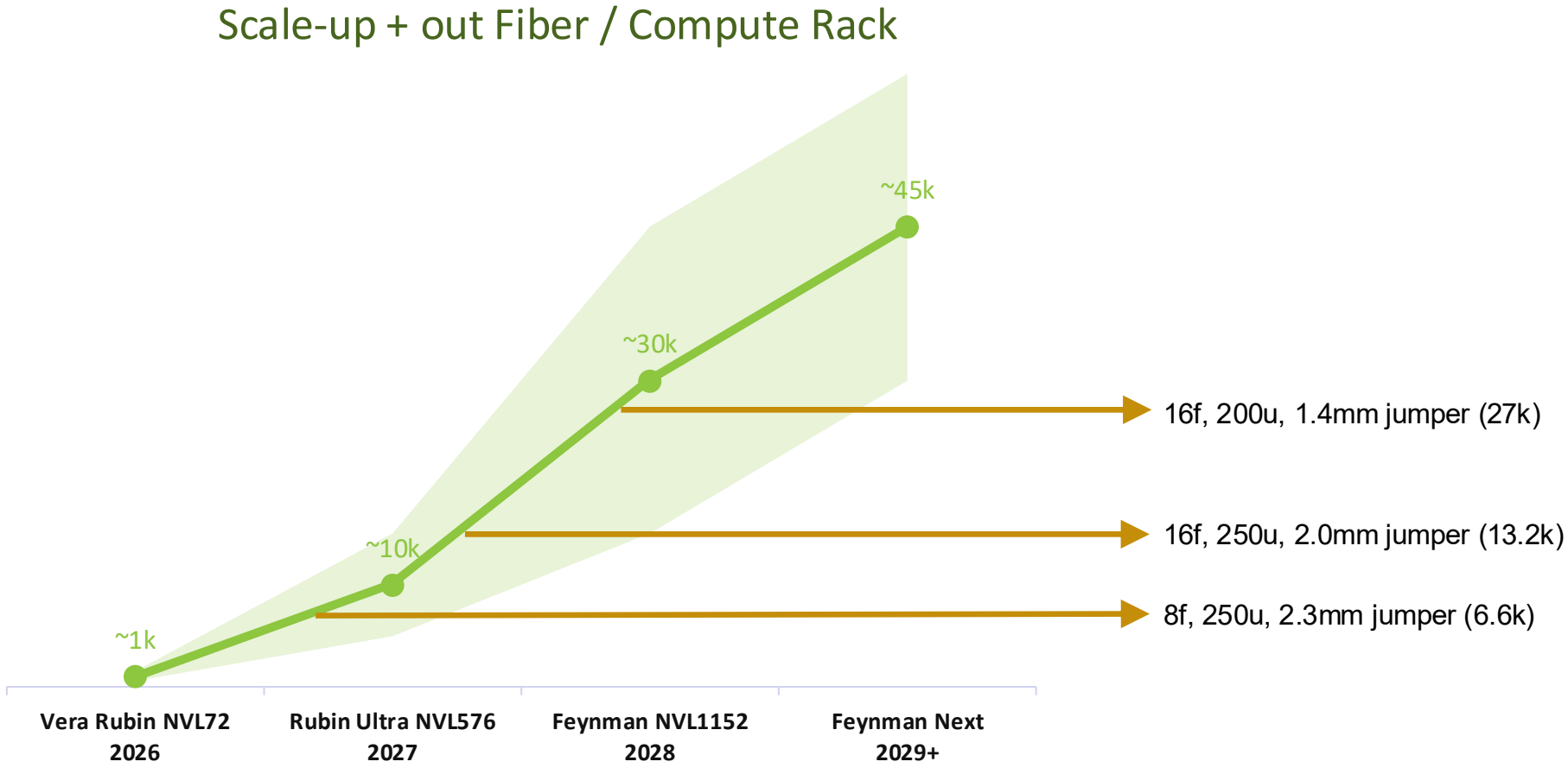


Architecture choices change the curve, but not the direction



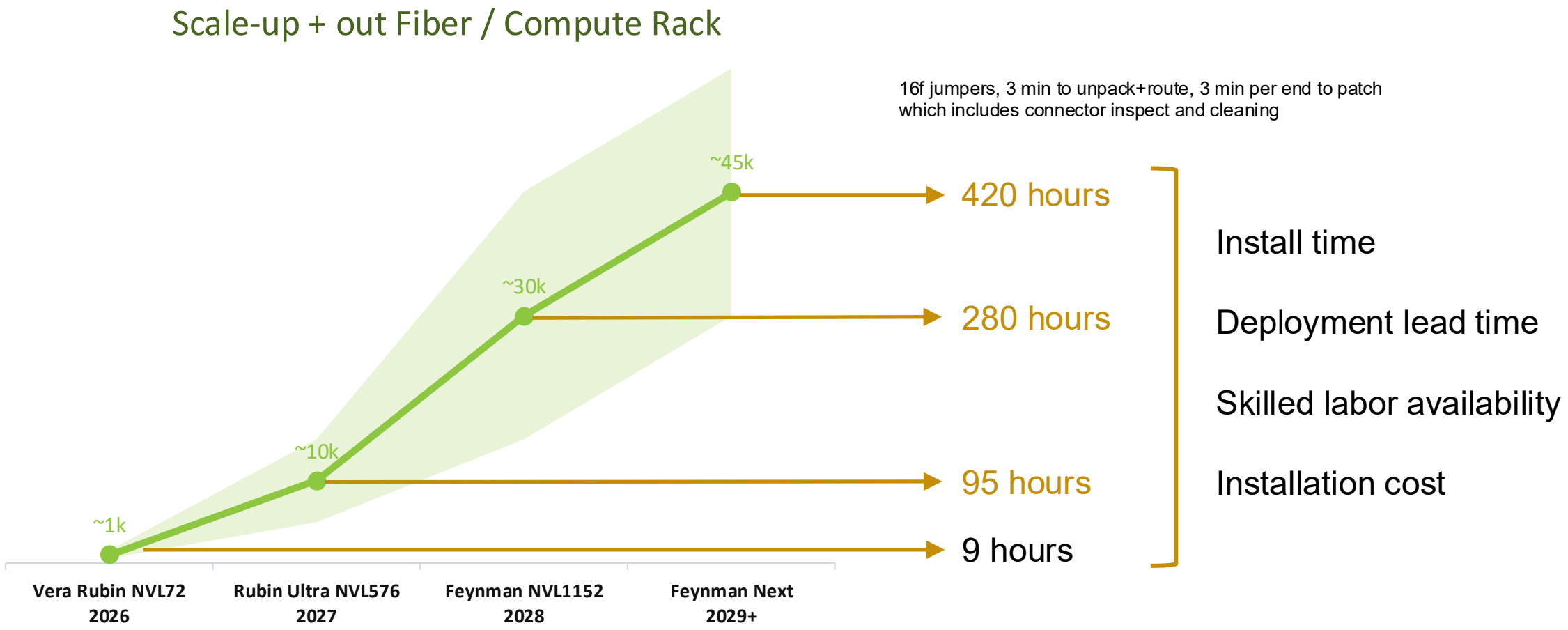
Could vary by:
- GPU/rack
- SERDES rate
- λ usage

Conventional components become space limiting

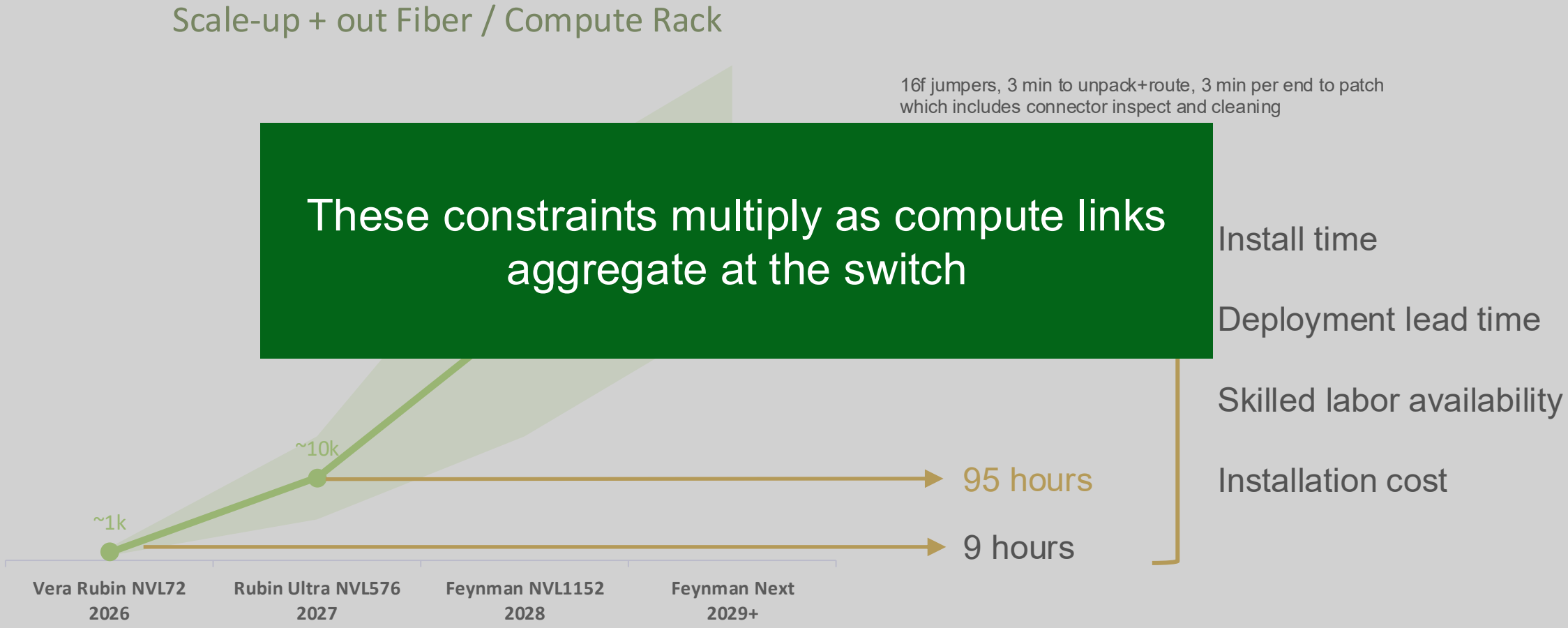


Assuming MGX rack design and typical routing

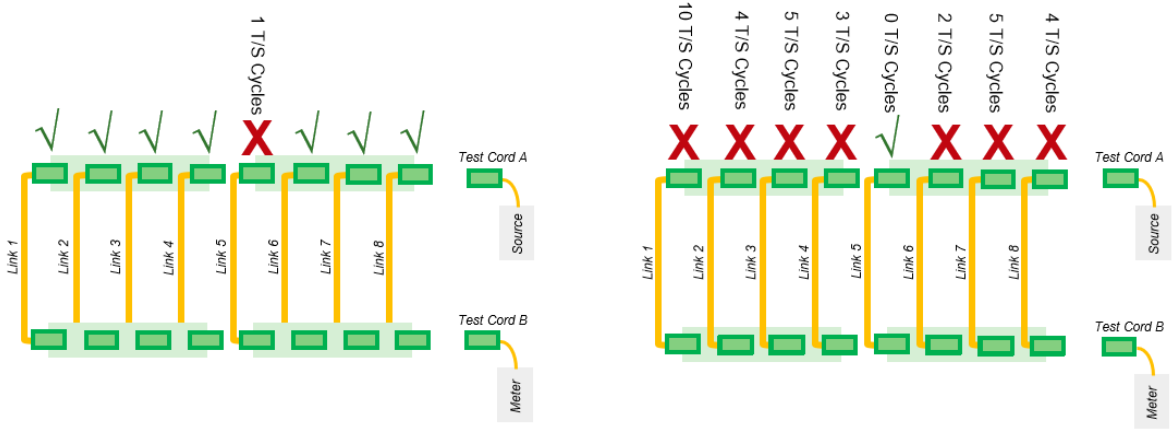
Conventional installation becomes labor-limiting



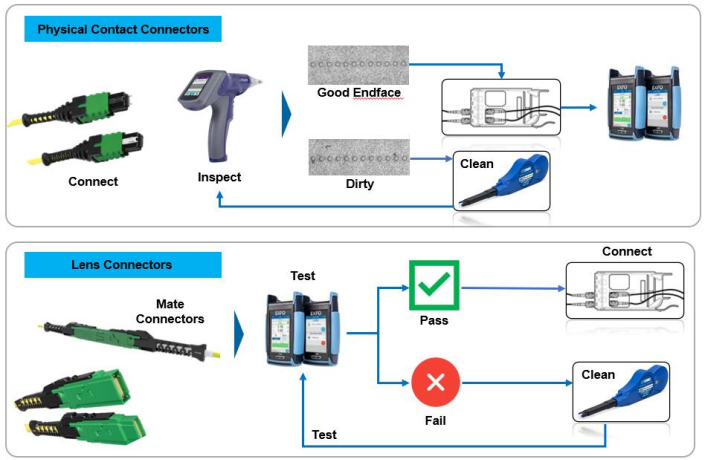
Switch Aggregation Multiplies Both Constraints



Operational Advantage:
Speed of Deployment



Connector	Time to Test, T/S and Retest	Time per rack
MMC with Prizm	8 min. 43 secs.	9.3 hours / rack
MMC, Physical Contact	1 hour 19 min. 23 secs.	84.7 hours / rack



- 16 Port Deployment Time Trial
  - Prizm 89% Faster Deployment vs Physical Contact
- Need faster deployment
  - Less time cleaning
  - Less time troubleshooting
- VSFF Mass Insertion
  - More lenses, fewer plug ins

Why Lens? Physical Advantage
Remate Stability without Cleaning

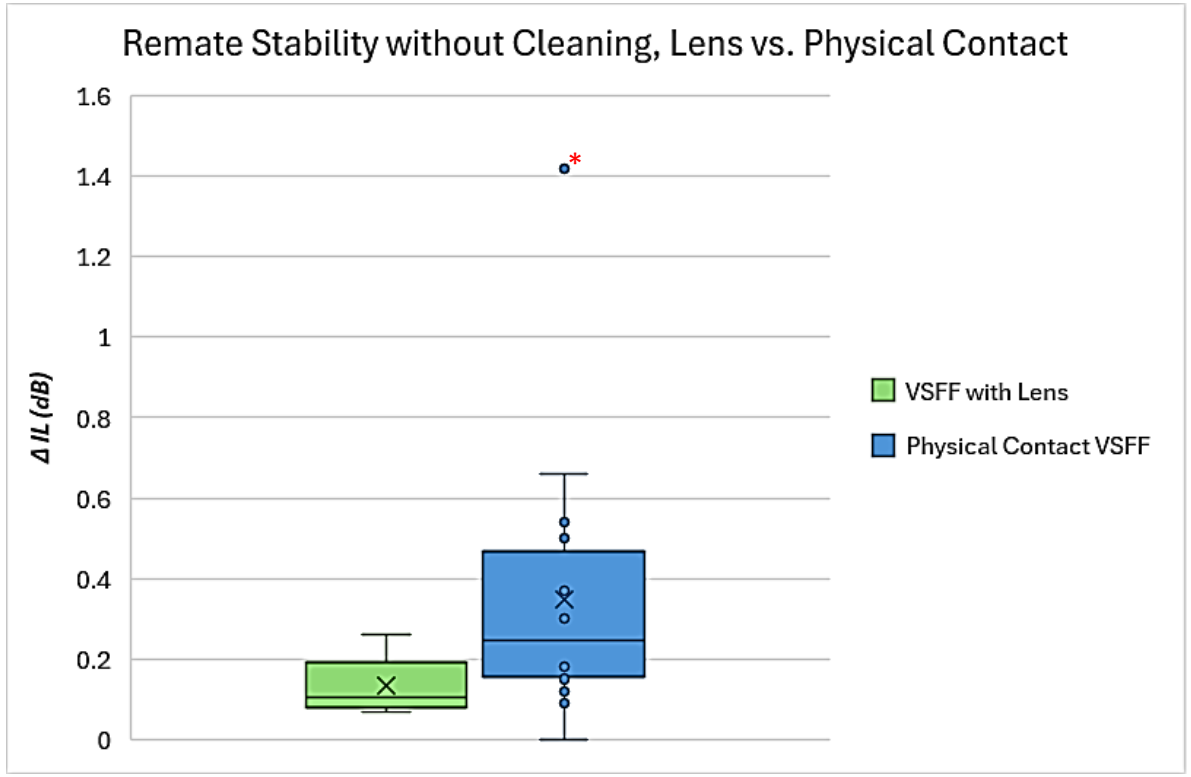
EXFO FTBx-88480 Series

Hisense 800G 2DR4 transceiver

System Under Test: 5 Mated Connector Pairs

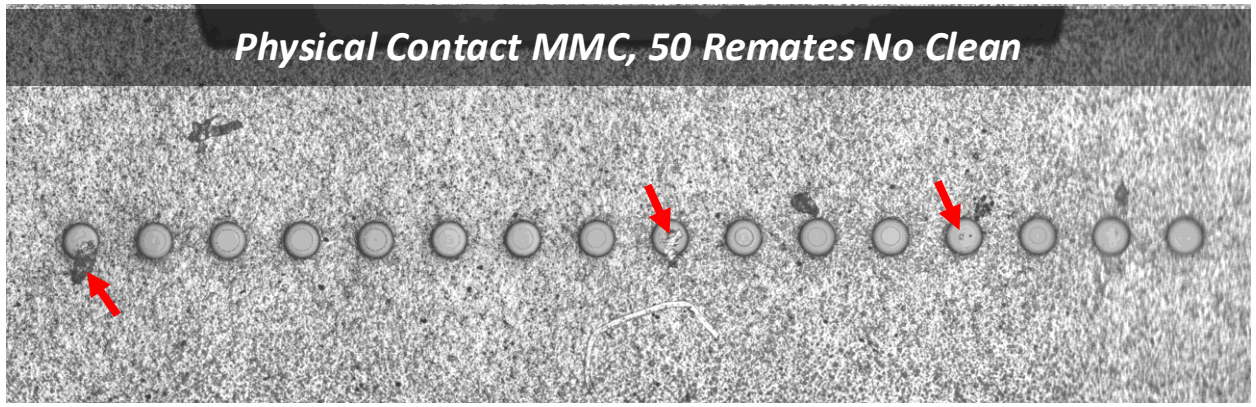
- No cleaning
- Scope, Mate, Measure at:
  - 1 Mating
  - 10 Matings
  - 50 Matings

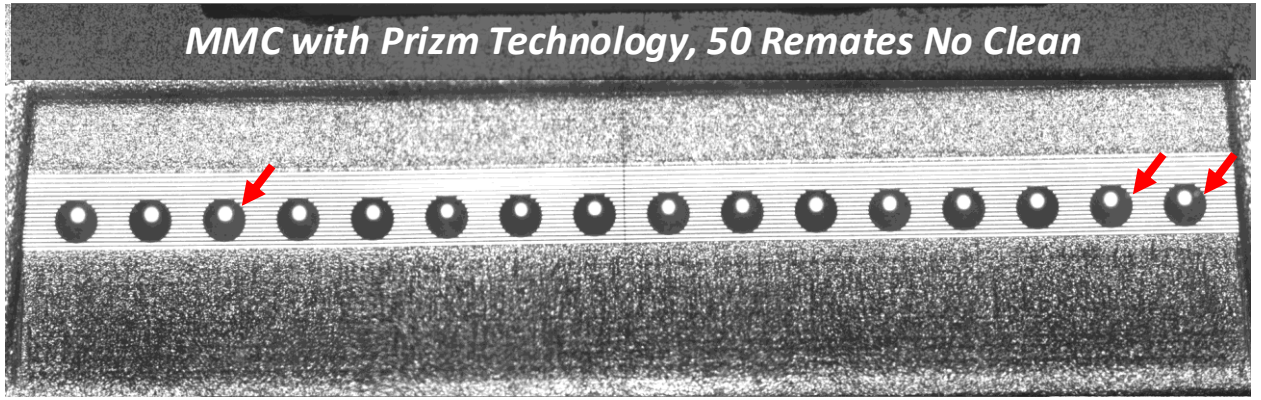
Stable lens link performance without cleaning



* 2027 MMC Fundamental Understanding Completed

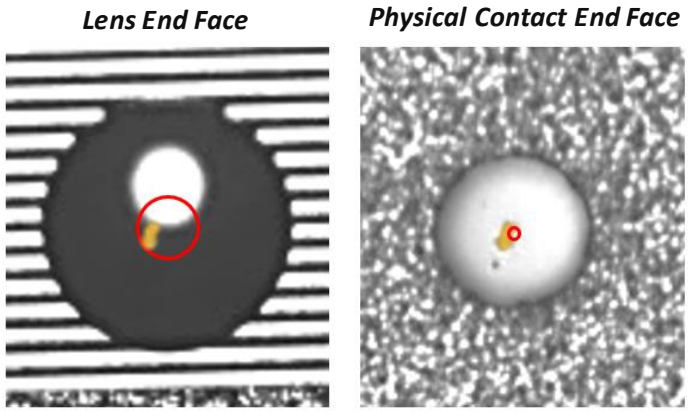
Why Lens: Physical Advantage
Debris Resilient





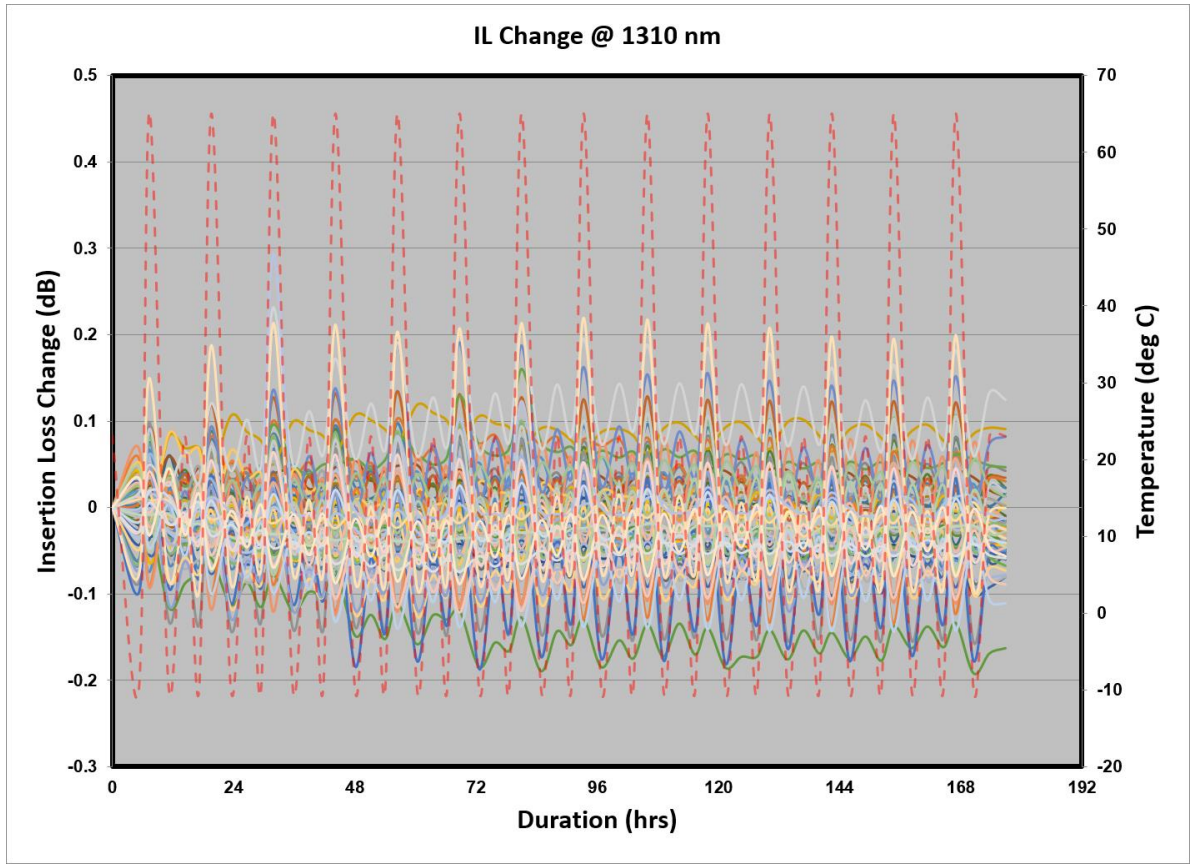
Debris generated on physical contact (top image) had more significant IL impact than debris generated on Lens (bottom image).

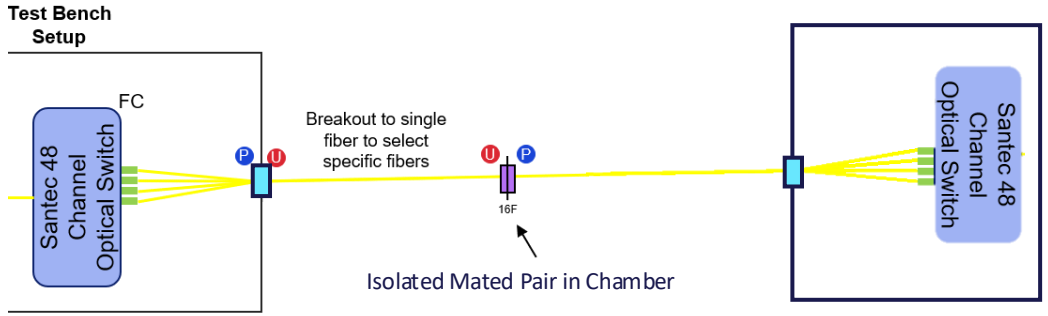
- Decreased IL impact due to debris creates stable link
- *5 mated pairs*, 50 remates, **no cleaning**
- Physical contact Max ΔIL = 1.42 dB
- Lens Max Δ IL = 0.26 dB



Difference in core coverage by yellow particle:
15% Lens core vs 100% Physical Contact core

Why Lens: Physical Advantage
Condensation Resilient



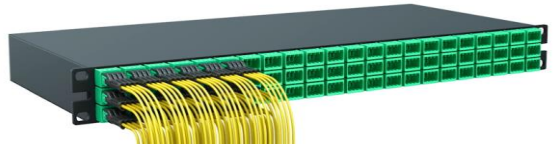


- 5 mated pairs, Prizm passed maximum IL spec for GR-1435 4.4.4: Condensation testing
- Max Temp = 65° C
- Max Humidity = 95% RH
- Max Δ IL at 1310 nm = 0.29 dB
  - GR-1435 4.4.4 Max Δ IL Spec is <0.3 dB during test

Operational Advantage:
Density & Deployment



4,224 fibers in 1RU using 16f MMC connectors



8,448 fibers in 1RU using 32f MMC connectors

Denser Solutions, Faster Deployment

Base 16F Design

32F on roadmap

- Same ferrule, same form factor
- Enables 3-6x front panel density

30% mating force of Physical Contact

- Rise of Blindmate solutions

94% reduction in cleaning materials

16f Ferrule



32f Ferrule





96f (6x 16f)

Mass Insertion at Patch Panel



192f (6x 32f)



64f (4x 16f)

Optical Blindmate / Backplane



128f (4x 32f)



192f (12x 16f ferrules)



384f (12x 32f ferrules)

NOW → NEXT

Manufacturing Scale Advantage

Passive Fiber Alignment

Passive mechanical alignment features enable scaling a termination process that requires simple / no automation



Speed to Scale

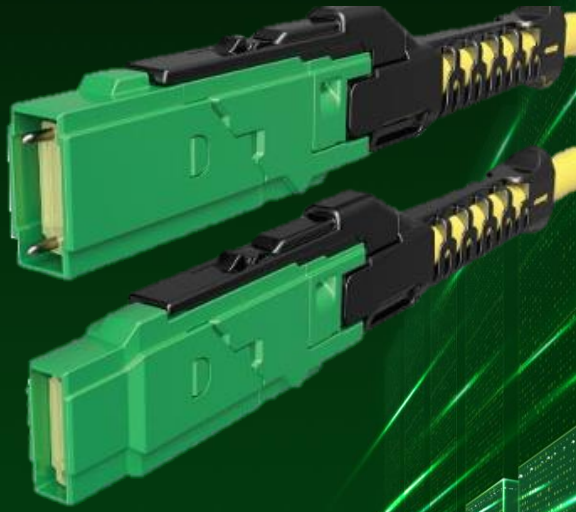
MMC with Prizm Technology demonstrates a repeatable manufacturing process that scales to AI network volumes

90 days from lab trial to manufacturing deployment

2 days to convert existing physical contact MMC line to Lens

Connector Performance: *Path to 0.5 dB*

Bringing It Together



- **Field Ready Connector** – Debris resilient, Remate Stable, 89% Faster Deployment
- **Predictability** – 81% lower max Δ IL than physical contact across 50 remates, no cleaning
- **Density** – 3-6x higher front panel, 2x higher blind mate with no connector redesign
- **Manufacturing Scalability** – Passive alignment enables repeatable, low-automation termination and rapid scale-up to AI-network production volumes.

MMC with Prizm Technology is field-ready today for the AI-scale networks of tomorrow.

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